ENCLOSURE II

Department of Information Technology

Delhi Technological University

IT401 – B. Tech Project-I

Project Synopsis Format

Team Details

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Supervisor:

Dr. Bindu Verma

1. Title of Project

GenoVista: Disease causing Genomic Variant Interpretation using AI-ML

2. Introduction

Genomic variant interpretation is a cornerstone of precision medicine, particularly in cancer research and genetic disorders. Variants such as single nucleotide substitutions can be either benign or pathogenic, depending on whether they disrupt critical biological functions. While curated databases like **ClinVar** provide clinical classifications, many variants remain uncharacterized, limiting their utility for diagnostics and research.

Stanford's recently released **Evo 2 model**—a large genomic foundation model—can predict the functional impact of mutations by analyzing sequence plausibility. This project proposes to integrate Evo 2 with ClinVar in a **full-stack web application** that enables users to query variants, compare predictions, and assess pathogenicity. The platform will be powered by **Modal serverless GPUs for Evo 2 inference**, a **FastAPI backend in Python**, and a **Next.js frontend** with a modern UI.

3. Problem Statement

Despite advances in genomics, there is no accessible tool that combines cutting-edge AI predictions with trusted clinical databases in a **user-friendly platform**. Many genetic variants, especially in cancer-related genes like **BRCA1**, are left as Variants of Uncertain Significance (VUS). Researchers and students lack an integrated system that allows them to explore pathogenicity predictions, compare with curated data, and gain insights into disease risk.

4. Motivation

With the rising importance of **genomic medicine**, tools that improve **variant interpretation** are highly relevant. Evo 2's ability to predict variant functional effects offers a chance to triage uncertain variants computationally. By combining Evo 2 predictions with ClinVar's authoritative classifications, this project delivers a system that is valuable for **research, teaching, and potential pre-clinical exploration**. Moreover, deploying it with **serverless GPUs (Modal)** ensures scalability and accessibility for real-world users.

5. Literature Review

1. Evo 2 (2025)
    - Evo 2 is a next-generation, AI-powered model trained on billions of genomic sequences. It achieves high accuracy in predicting the effects of genetic variants—including rare or novel ones—by leveraging evolutionary constraints, outperforming many prior tools in coding and noncoding variant assessment.
2. Machine Learning for Variant Classification (2022–2025)
    - Recent work applies machine learning, such as penalized logistic regression, to classify variants using ACMG/AMP guidelines and feature annotation. These approaches increase accuracy in identifying pathogenic variants, especially for those previously labeled as "uncertain significance".

3. Advances in Clinical Databases: ClinVar (2022–2025)
    - Ongoing updates to ClinVar offer better curation, variant filtering, and integration with clinical workflows. These improvements have made ClinVar essential for both automated and expert-driven variant analysis, supporting precision medicine and large-scale genomics.

4. Comprehensive Variant Analysis Platforms (2024–2025)

- Platforms like DRAGEN and omnomicsNGS combine genome-scale analysis, cloud computing, and AI prediction tools. They automate variant annotation, support clinical standards, and handle a variety of variant types, making high-throughput and robust variant interpretation more accessible.

6. Expected Goal

- TheEvo 2 model for variant effect prediction
- Predict pathogenicity of single nucleotide variants (pathogenic/benign)
- Comparison view: ClinVar classification vs Evo 2 prediction
- Prediction confidence estimation
- Genome assembly selector (e.g., hg38)
- Gene selection via chromosome browsing or search (e.g., BRCA1)
- Access to full reference genome sequence (via UCSC API)
- Gene and variant data exploration (via NCBI ClinVar/E-utilities)
- Backend in Python with FastAPI
- GPU-accelerated Evo 2 inference via Modal (H100 serverless GPUs)
- Responsive Next.js web app frontend
- Modern UI with Tailwind CSS & Shadcn UI

Expected Outcomes:

- A working **prototype application** for variant effect prediction and pathogenicity comparison.
- A curated dataset of variants scored by Evo 2 for benchmarking.
- Potential publication in a bioinformatics/AI conference on integrating AI with clinical genomics.

25/8/25

Name & Signature of all Team Members
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